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A new active learning strategy for soft sensor modeling based on feature reconstruction and uncertainty evaluation

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ABSTRACT

Soft sensor techniques have been increasingly applied in chemical processes to predict key variables online. Traditionally, soft sensors modeling needs labeled training samples that contain both subsidiary and key variables. However, labeled samples are limited because key variables are always difficult to be obtained online. Thus, a novel active learning (*AL*) strategy is proposed to reduce the labeling cost, which iteratively select the most informative candidates by jointly evaluating two criteria: (i) representativeness and (ii) uncertainty. The representative samples are defined as those whose features have the best reconstruction of global features extracted by kernel principal component analysis (*KPCA*), while the uncertain samples can be selected by the estimated variance based on Gaussian process regression (*GPR*) model. Then, the optimization scheme is also introduced to solve the optimization problem derived by the two sampling criteria. Three industrial application case studies show that the proposed *AL* strategy exhibits a good capability to select the most informative samples, which can improve the performance of the soft sensor.

1. Introduction

Recently, soft sensor models [1,2]are widely used to predict key variables that are difficult to measure online in industrial processes. The soft sensors are mainly divided into two categories: model-driven [3] and data-driven [1]. Compared with the former, data-driven soft sensors are gaining popularity due to their independence from a priori knowledge. Generally, data-driven soft sensors always need labeled samples that contain both subsidiary and key variables [4–6] for modeling. However, the acquirement of key variables (such as butane concentration and flue gas oxygen content) is costly due to the expensive instruments or offline analysis by human efforts. As a result, we usually have infrequent or irregular labeled samples [7], but plenty of unlabeled samples that only contain subsidiary variables. How to leverage the abundant unlabeled samples with a few labeled samples to help supervised learning is a challenging problem during the past few years. One of the effective solutions to this problem is semi-supervised learning [8,9]which addresses methods for automatically exploiting unlabeled samples in addition to labeled samples to improve learning performance. Another solution is active learning (*AL*) [10] which reduces the labeling efforts by obtaining the labels of the most informative samples. Thus, instead of processing data passively, the active learner performs an intervention over the sample space to control what samples are added to the training dataset. In this way, we expect that the soft sensor model can achieve good performance using as few labeled samples as possible.

In terms of query formation, there are three categories of *AL* in the literature: (1) query synthesis [11], ("query" means obtaining the real labels by offline analysis for soft sensor research), (2) stream-based sampling [12]and (3) pool-based sampling [13]. In Query synthesis strategy, the candidates (training samples that only contain subsidiary variables) are not only collected from historical database, but also synthesized from the sample feature space. Thus, some candidates may have no obvious physical meaning, which are hardly labeled by offline analysis. The steam-based sampling strategy selects one sample at a time and decides whether to query it or not, which is only appropriate in situations where memory or processing power is limited. As a result, most efforts have focused on pool-based sampling in recent years. In general, a pool-based sampling *AL* algorithm aims to iteratively select the informative samples to enlarge the training dataset. At each

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iteration, it evaluates and ranks the entire collection of candidates in the pool. Then, the real labels are assigned to the selected samples and added into the training dataset. Finally, the desired model should be updated by training the new dataset. The process is iterated until the desired performance is met. Some previous pool-based *AL* algorithms design selection mechanism to select the representative samples [14] from the pool that consists of the unlabeled candidates. However, these algorithms are all label independent, that is, the label information obtained at the current iteration are always ignored at the following iteration. Other previous work usually need a number of pregiven labeled samples [15] to train a model and then select the candidates that are most uncertain to predict. Nevertheless, the number of labeled samples is always extremely sparse during the initial several iterations so that the desired model can not be built directly, which may lead to inappropriate sampling criterion. All these drawbacks motivate this paper.

In this paper, we propose a novel *AL* soft sensing framework, called *AL-RU* which means active learning strategy by jointly selecting representative and uncertainty samples for labeling. First, we adopts feature reconstruction method based on kernel principal component analysis (*KPCA*) [16]to select the representative samples from the unlabeled candidate pool. Then, those selected samples are labeled by human efforts and added into the training dataset. Next, we adopt the gaussian process regression (*GPR*) [15,17] method to build the soft sensor model and calculate the estimated variance of the candidates. The uncertainty of each candidate can be evaluated by its estimated variance, which can be considered as the other sampling criterion at the following iteration. Finally, a computational scheme is introduced to solve the optimization problem derived by the two sampling criteria.

The remaining parts of this paper are organized as follows. Section 2 describes the procedures of *AL* strategy for selecting the representative samples. In section 3, we adopt *GPR* method to design the other sampling criterion for selecting the uncertain samples. In section 4, we will give the framework of *AL-RU* and propose the optimization scheme to solve the optimization problems. Three case studies of soft sensor modeling are carried out in section 5. Finally, the conclusion and plans for future work are discussed.

2. AL strategy based on feature reconstruction

The main issue with selecting the representative samples is how to generate a query to evaluate the informativeness [14] only based on the unlabeled candidate samples. In soft sensor modeling, the unlabeled candidate samples that consist of subsidiary variables frequently have nonlinear and strong coupling characteristics. *KPCA*, based on kernel tricks, are widely used to extract these characteristics, which maps all the samples from their original space to a higher dimensional feature space where eigenvalue decomposition is performed. As a result, we can extract global features of the candidates by *KPCA*. Then, the most representative samples can be naturally defined as those whose features have the best reconstruction of global features. Inspired by this, we propose the *AL* strategy based on feature reconstruction, called *AL-R*, which adopts *KPCA* to select the representative samples.

2.1. Feature extraction based on KPCA

Given the candidate sample pool as $X = \{x_1, x_2, \ldots x_n\} \in \mathbb{R}^{m\times n}$, where n denotes the number of the unlabeled samples and m is the number of the subsidiary variables. Here, each candidate sample $x_i \in X$ is projected onto a high-dimensional space ($\mathbb{F}$), which can be expressed as $\Phi : \mathbb{R}\rightarrow\mathbb{F}$. The covariance matrix can be formulated as:

$$C^{\mathbb{F}} = \frac{1}{n-1}\sum_{i=1}^{n}\phi(x_i)(\phi(x_i))^T \tag{1}$$

The sample input vectors $\phi(x_i)$ should be centered as $\sum_{i=1}^{n}\phi(x_i) = 0$ .Then, we should find the eigenvalue α_i and eigenvector $\boldsymbol{p}_i$ of $C^{\mathbb{F}}$ by solving the following eigenvalue problem:

$$\alpha_i \boldsymbol{p}_i = C^{\mathbb{F}}\boldsymbol{p}_i \tag{2}$$

Note that nonlinear mapping ϕ is usually defined as a kernel function. Thus, Eq. (2) can be transformed to an eigenvalue problem (3) using kernel tricks.

$$\gamma_i \boldsymbol{v}_i = K_c \boldsymbol{v}_i \quad i = 1, 2, \ldots n \tag{3}$$

where K_c is the centered matrix which is derived from the following equations:

$$K_c = K - JK - KJ + JKJ \tag{4}$$

where K is the $n \times n$ kernel matrix, and J is the $n \times n$ matrix with each element is 1. Any function satisfying Mercer's condition [18] can be applied as K. In this paper, radial basis function (*RBF*) $k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{1}{2\sigma_n^2}(x_i - x_j)^2\right)$ is used as the kernel function. The terms σ_f^2 and σ_n correspond to signal variance and width parameter. Here, the k highest eigenvalues $\lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_k\}$ and the corresponding eigenvectors $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k]$ of K_c can be calculated by the eigenvalue decomposition method. Here, k principal components can be sufficient to represent the most relevant information contained in the centered kernel matrix K_c, which should minimize the objective function $\left\|\sum_{i=1}^{k}\lambda_i / \sum_{i=1}^{n}\lambda_i - \eta\right\|_2$ [19]. Here, η is a variable parameter, which is determined by the users. In this paper, we set η to 0.85 according to the engineering experience. The principal eigenvector matrix U and the corresponding eigenvalues λ are presented as the global features of the candidate samples in the pool.

2.2. Sampling criterion for selecting the representative samples

In *AL-R* strategy, only a fraction of candidate samples can obtain their labels at each iteration. More specifically, it is necessary to calculate a selection matrix Λ. The selected sample set X_s can be computed by $X_s = X\Lambda$. In the kernel space $\mathbb{F}$, the selected sample centered kernel matrix can be computed by:

$$K_{sc} = K_c\Lambda = K_c(X, X_s) \tag{5}$$

Here, $X_s = \{x_{s_1}, x_{s_2}, \ldots, x_{s_k}\}$ denotes the selected ID set which contains the informative samples selected by *AL* algorithms and $\{s_1, s_2, \ldots, s_k\}$ means the set of the selected ID number. If the selected sample set X_s is labeled and added into the training set, the soft sensor model can obtain the most precise predictive value. $\Lambda \in \mathbb{R}^{n\times k}$ is a special matrix which satisfies two condition: (1) each column contains a single 1 with 0s everywhere else; and (2) all these 1s are in different rows. If a row contains a single 1, the row number is the selected ID number.

Here, the k highest eigenvalues $\boldsymbol{\theta} = \{\theta_1, \theta_2, \ldots, \theta_k\}$ and the corresponding eigenvectors $Q = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_k]$ of K_{sc} can also be calculated by the eigenvalue decomposition method. In the kernel space $\mathbb{F}$, the reconstructed samples $X' = \{x'_1, x'_2, \ldots, x'_n\}$ can be computed by the following equation:

$$X' = U^T K_c \tag{6}$$

The covariance matrix of X' can be calculated by Eq. (7):

$$C' = \frac{1}{n-1}X'X'^T = \frac{1}{n-1}U^T K_c K_c^T U = diag\{\lambda_1, \lambda_2, \ldots, \lambda_k\} \tag{7}$$

Due to the pairwise orthogonal characteristic of the eigenvectors $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k]$, the covariance matrix C' has two properties: (1) all

entries on the primary diagonal are the k highest eigenvalues in descending order, and (2) the absolute values of other entries are approximately equal to zero.

Similarly, the covariance matrix C'_s of the selected samples $X'_s = X'\Lambda$ can be formulated as:

$$C'_s = \frac{1}{n-1} U^T K_c \Lambda \Lambda^T K_c^T U = diag\{\theta_1, \theta_2, \ldots, \theta_k\} \tag{8}$$

To ensure that the features of the selected samples are the best reconstruction of global features, it is necessary to make the covariance matrice C'_s and C' as equal as possible. Thus, the representative samples are naturally defined as those which minimize the objective function in the optimization problem (16).

$$\begin{aligned} \min \quad & \sum_{i=1}^{k} \|\theta_i - \lambda_i\|^2 \quad + \mu_1 \left\| \left| \sum_{i=1}^{k} \sum_{j=1}^{k} c_{ij} \right| - tr(C'_s) \right\|^2 \\ s.t. \quad & \begin{cases} C'_s = \dfrac{1}{n-1} U^T K_c A K_c^T U \\ C'_s = \{c_{ij}\}_{i,j \in \Omega, \Omega = 1,2,\ldots,k} \\ \theta_i = \{c_{ii}\}_{i=1,2,\ldots k} \\ A = \Lambda\Lambda^T = diag\{a_{11}, a_{22}, \ldots, a_{nn}\} \\ K_c = K - JK - KJ + JKJ \\ a_{ii} = \{0,1\}, i = 1,2,\ldots n \\ \sum_{i=1}^{n} a_{ii} = L \\ J = \dfrac{1}{n} ones(n,n) \end{cases} \end{aligned} \tag{9}$$

where A is a diagonal matrix whose diagonal entries $\{a_{ii}\}_{i=1,2,\ldots,n}$ are optimization variables. L denotes the number of selected samples at each iteration, and μ_1 is a suitable constant. Given the optimal solution $\widehat{A}$ of (16), we select those samples whose corresponding entries in the diagonal matrix are 1. After obtaining the labels of the selected samples, we can use any supervised learning algorithm to train the soft sensor model. Problem (16) is not a standard optimization problem, thus we will give the computational scheme to solve it in section 4.1.

3. *AL* strategy based on uncertainty evaluation

In section 2, the sampling criterion based on feature reconstruction is applied to select the representative samples which should query their labels by human efforts. Then, the soft sensor model can be built by the new labeled samples. How to select the modeling tool will be discussed. First, we have extracted the nonlinear features from the candidates by *KPCA*. As a result, the nonlinear modeling tool should be considered. Next, the estimated variance of the predicted values should be easily given by the nonlinear modeling tool, which is used to design the *AL-U* strategy. Finally, the kernel parameters used in *AL-R* strategy should be updated by use of the new labeled samples. To sum up, *GPR* is the powerful machine learning technique, which can meet these conditions. Because *GPR* is a nonlinear modeling tool, which adopts kernel tricks and updates the kernel parameters at each iteration. Meanwhile, the estimated variance of the predicted values can be easily calculated by the kernel function, which can be used to design the *AL-U* strategy. Thus, using *GPR* soft sensor model to design the *AL-U* strategy for selecting the most uncertain samples will be detailed as follows.

3.1. Estimated variance based on GPR

Gaussian process (*GP*) is a collection of random variables with joint multivariate Gaussian distribution. Given a set of L labeled training samples $\{x_i, y_i\}_{i=1,2,\ldots,L}$, where $Y = \{y_i\}_{i=1,2,\ldots,L}$ denotes the real labels obtained by human efforts and $X = \{x_i\}_{i=1,2,\ldots,L}$ denotes the selected samples. *GPR* is defined so that the regression function $f(X)$ has a Gaussian prior distribution over function $f(X) \sim GP(0, K(X,X) + \delta^2 I)$ with zero mean and variance δ^2. $K(X,X)$ is a covariance matrix with each element defined by the widely used function $k(x_i, x_j) = \sigma_f^2 \exp\left(-\frac{1}{2\sigma_n^2}(x_i - x_j)^2 \right)$. The covariance function is parameterized by $\varphi = \{\sigma_f, \sigma_n\}$, which is called "hyper-parameter".

Given a testing sample x_* from the candidate sample pool, *GPR* can find the predictive output $f(x_*)$ with probabilistic confidence level. The joint distribution of the training and testing samples is still a Gaussian distribution:

$$P(f(X) f(x_*)) \sim GP\left(0, \begin{bmatrix} k(X,X) + \delta^2 I & k(X,x_*) \\ k(x_*,X) & k(x_*,x_*) \end{bmatrix}\right) \tag{10}$$

where $k(X,x_*)$ and $k(x_*,X)$ denote the covariance matrix between the training and testing samples, and $k(x_*,x_*)$ denotes the covariance matrix of the testing sample. Conditioning the joint distribution yields the predicted mean value $f(x_*)$ with the corresponding variance $var(x_*)$:

$$\begin{aligned} f(x_*) &= k(x_*,X)\left(K(X,X) + \delta^2 I\right)^{-1} y = k(x_*,X)\Theta \\ \mathrm{var}(x_*) &= k(x_*,x_*) - k(x_*,X)\left(K(X,X) + \delta^2 I\right)^{-1} k(X,x_*) \end{aligned} \tag{11}$$

The hyper-parameter can be estimated by maximizing the logarithm of the likelihood function: $f_{likehood} = \log p(f(x_*)|\varphi, X) = -\frac{1}{2} f(x_*)^T K(X,X)^{-1} f(x_*) - \frac{1}{2}\log\left|K(X,X)\right| - \frac{n}{2}\log 2\pi$. Using standard optimization methods, such as the conjugate gradient algorithm, the estimated value of the hyper-parameter φ can be obtained. Here, the estimated variance of the predicted value $f(x_*)$ can be calculated by Eq. (11).

3.2. Sampling criterion for selecting the uncertain samples

Based on *GPR* soft sensor model, the estimated reliability of each candidate sample can be measured by its estimated variance. If the candidate sample has a large estimated variance, we can say it has a significant estimation uncertainty which shows more informative to the *GPR* model. Thus, the candidate samples with large estimated variance should has the high priority for selection, which can be used to design the other sampling criterion. As with *AL-R* strategy, we also introduce the selection matrix $\Lambda \in \mathbb{R}^{n \times k}$. The sum of estimated variance of the selected samples can be calculated by:

$$\begin{aligned} \sum \mathrm{var}(X\Lambda) = tr\Big(& \Lambda^T K(X,X) \Lambda \\ & - \Lambda^T K(X,X)\left(K(X,X) + \delta^2 I\right)^{-1} K(X,X)\Lambda \Big) \end{aligned} \tag{12}$$

The optimal selection matrix $\widehat{\Lambda}$ is defined to make Eq. (12) obtain its maximum. Thus, the most uncertain samples are naturally defined as those which maximize the objective function in the optimization problem (20)

$$\max \quad tr\left(\Lambda^T K(X,X)\Lambda - \Lambda^T K(X,X)\left(K(X,X)+\delta^2 I\right)^{-1} K(X,X)\Lambda\right)$$
$$s.t.\begin{cases} A = \Lambda\Lambda^T = diag\{a_{11}, a_{22}, \ldots, a_{nn}\} \\ a_{ii} = \{0,1\}, i = 1,2,\ldots,n \\ \sum_{i=1}^{n} a_{ii} = L \end{cases} \tag{13}$$

where $\{a_{ii}\}_{i=1,2,\ldots,n}$ is the set of optimization variables. Once the optimal solution $\widehat{A}$ is obtained, the most uncertain samples are defined as those whose corresponding entries of $\widehat{A}$ are 1. The computational scheme of solving problem (20) will be given in section 4.1.

4. *AL* strategy based on feature reconstruction and uncertainty evaluation

In this section, we will discuss a novel *AL* soft sensing framework (*AL-RU*) that combines feature reconstruction and uncertainty evaluation. At the first iteration of *AL-RU* strategy, we only adopt *AL-R* strategy to select the most representative samples due to the lack of the label information. Then, the real labels are assigned to the selected samples for modeling. Using the label information, we can train a *GPR* model and identify the candidates that are most hard to be predicted by *AL-U* sampling criterion which can be considered as the supplement to *AL-R* sampling criterion. As a result, the optimization problem (21) derived by *AL-RU* strategy should contain two optimization indexes: representativeness and uncertainty, which is detailed as follows:

$$\min \quad \left(\sum_{i=1}^{k}\|\theta_i - \lambda_i\|^2 + \mu_1\left\|\left|\sum_{i=1}^{k}\sum_{j=1}^{k} c_{ij}\right| - tr(C'_s)\right\|^2\right)$$
$$-\mu_2 tr\left(\Lambda^T K\Lambda - \Lambda^T K\left(K+\delta^2 I\right)^{-1} K\Lambda\right)$$
$$s.t.\begin{cases} C'_s = \dfrac{1}{n-1}U^T K_c A K_c^T U \\ C'_s = \left\{c_{ij}\right\}_{i,j\in\Omega,\Omega=\{1,2,\ldots,k\}} \\ \theta_i = \left\{c_{ij}\right\}_{i=1,2,\ldots,k} \\ A = \Lambda\Lambda^T = diag\{a_{11}, a_{22}, \ldots, a_{nn}\} \\ a_{ii} = \{0,1\}, i = 1,2,\ldots,n \\ \sum_{i=1}^{n} a_{ii} = L \\ K_c = K - JK - KJ + JKJ \\ J = \dfrac{1}{n}ones(n,n) \end{cases} \tag{14}$$

The first two terms of the objective function in (21) denotes the *AL-R* strategy based on feature reconstruction, while the third term stands for the *AL-U* strategy by evaluating sample uncertainty. Here, μ_1 and μ_2 are the regularization parameters. We will give the optimization scheme to solve the optimization problem (21).

4.1. Optimization scheme

In this section, we will develop the optimization scheme to solve (16). As observed, $\{a_{ii}\}_{i=1,2,\ldots,n}$ are the optimization variables in (16). The objective function can be explicit rewritten as a function of $\{a_{ii}\}_{i=1,2,\ldots,n}$. First, we will investigate the type of the optimization problem to which (16) belongs. The covariance matrix C'_s can be expressed as:

$$C'_s = \frac{1}{n-1}U^T K_c \Lambda\Lambda^T K_c^T U$$
$$= \frac{1}{n-1}\begin{bmatrix}\boldsymbol{u}_1^T \\ \boldsymbol{u}_2^T \\ \vdots \\ \boldsymbol{u}_k^T\end{bmatrix}\begin{bmatrix}K_{c1} & K_{c2} & \cdots & K_{cn}\end{bmatrix}\begin{bmatrix}a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn}\end{bmatrix}\begin{bmatrix}K_{c1}^T \\ K_{c2}^T \\ \vdots \\ K_{cn}^T\end{bmatrix}\begin{bmatrix}\boldsymbol{u}_1^T \\ \boldsymbol{u}_2^T \\ \vdots \\ \boldsymbol{u}_k^T\end{bmatrix}^T$$
$$= \frac{1}{n-1}\begin{bmatrix}\sum_{i=1}^{n}\boldsymbol{u}_1^T K_{ci}K_{ci}^T\boldsymbol{u}_1 a_{ii} & \sum_{i=1}^{n}\boldsymbol{u}_1^T K_{ci}K_{ci}^T\boldsymbol{u}_2 a_{ii} & \cdots & \sum_{i=1}^{n}\boldsymbol{u}_1^T K_{ci}K_{ci}^T\boldsymbol{u}_k a_{ii} \\ \sum_{i=1}^{n}\boldsymbol{u}_2^T K_{ci}K_{ci}^T\boldsymbol{u}_1 a_{ii} & \sum_{i=1}^{n}\boldsymbol{u}_2^T K_{ci}K_{ci}^T\boldsymbol{u}_2 a_{ii} & \vdots & \sum_{i=1}^{n}\boldsymbol{u}_2^T K_{ci}K_{ci}^T\boldsymbol{u}_k a_{ii} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^{n}\boldsymbol{u}_k^T K_{ci}K_{ci}^T\boldsymbol{u}_1 a_{ii} & \sum_{i=1}^{n}\boldsymbol{u}_k^T K_{ci}K_{ci}^T\boldsymbol{u}_2 a_{ii} & \cdots & \sum_{i=1}^{n}\boldsymbol{u}_k^T K_{ci}K_{ci}^T\boldsymbol{u}_k a_{ii}\end{bmatrix} \tag{15}$$

where K_{ci} denotes the ith column of K_c. Then, the *AL-R* strategy requires that all entries on the primary diagonal of C'_s are the eigenvalues in descending order, while the absolute values of the other entries are approximately equal to zero. To guarantee the two properties above, the objective function becomes:

$$\min \sum_{p=1}^{k}\left\|\left(\sum_{i=1}^{n}\boldsymbol{u}_p^T K_{ci}K_{ci}^T\boldsymbol{u}_p a_{ii}\right) - \lambda_p\right\|^2 + \mu_1\left\|\sum_{p=1,q=1,p\neq q}^{k}\sum_{i=1}^{n}\boldsymbol{u}_p^T K_{ci}K_{ci}^T\boldsymbol{u}_q a_{ii}\right\|^2 \tag{16}$$

Next, by introducing an auxiliary variable $Z = [a_{11}, a_{22}, .., a_{nn}]^T$, the objective function (24) can be equivalently rewritten as:

$$\min_{a_{ii}} \frac{1}{2}Z^T GZ + Z^T F \tag{17}$$

where G is a Hessian matrix which can be expressed as:

$$G = \mu_1 MM^T + NN^T \tag{18}$$

Here,

$$M = \begin{bmatrix}\boldsymbol{u}_1^T K_{c1}K_{c1}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_1^T K_{c1}K_{c1}^T\boldsymbol{u}_k & \boldsymbol{u}_2^T K_{c1}K_{c1}^T\boldsymbol{u}_3 & \cdots \\ \boldsymbol{u}_1^T K_{c2}K_{c2}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_1^T K_{c2}K_{c2}^T\boldsymbol{u}_k & \boldsymbol{u}_2^T K_{c2}K_{c2}^T\boldsymbol{u}_3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \boldsymbol{u}_1^T K_{cn}K_{cn}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_1^T K_{cn}K_{cn}^T\boldsymbol{u}_k & \boldsymbol{u}_2^T K_{cn}K_{cn}^T\boldsymbol{u}_3 & \cdots \\ \boldsymbol{u}_2^T K_{c1}K_{c1}^T\boldsymbol{u}_k & \cdots & \boldsymbol{u}_{k-1}^T K_{c1}K_{c1}^T\boldsymbol{u}_{k-1} & & \\ \boldsymbol{u}_2^T K_{c2}K_{c2}^T\boldsymbol{u}_k & \cdots & \boldsymbol{u}_{k-1}^T K_{c2}K_{c2}^T\boldsymbol{u}_{k-1} & & \\ \vdots & \vdots & \vdots & & \\ \boldsymbol{u}_2^T K_{cn}K_{cn}^T\boldsymbol{u}_k & \cdots & \boldsymbol{u}_{k-1}^T K_{cn}K_{cn}^T\boldsymbol{u}_{k-1} & & \end{bmatrix}$$ and

$$N = \begin{bmatrix}\boldsymbol{u}_1^T K_{c1}K_{c1}^T\boldsymbol{u}_1 & \boldsymbol{u}_2^T K_{c1}K_{c1}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k^T K_{c1}K_{c1}^T\boldsymbol{u}_k \\ \boldsymbol{u}_1^T K_{c2}K_{c2}^T\boldsymbol{u}_1 & \boldsymbol{u}_2^T K_{c2}K_{c2}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k^T K_{c2}K_{c2}^T\boldsymbol{u}_k \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{u}_1^T K_{cn}K_{cn}^T\boldsymbol{u}_1 & \boldsymbol{u}_2^T K_{cn}K_{cn}^T\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_k^T K_{cn}K_{cn}^T\boldsymbol{u}_k\end{bmatrix}$$. Obviously, G is a positive semi-definite matrix and F is a vector which can be written as:

$$F = \begin{bmatrix}\sum_{i=1}^{k}u_i^T K_{c1}K_{c1}^T u_i\lambda_i & \sum_{i=1}^{k}u_i^T K_{c2}K_{c2}^T u_i\lambda_i & \cdots & \sum_{i=1}^{k}u_i^T K_{cn}K_{cn}^T u_i\lambda_i\end{bmatrix}^T \tag{19}$$

Finally, the optimization problem (16) can be formulated as:

$$\min_{a_{ii}} \frac{1}{2} \quad Z^T GZ + Z^T F$$
$$s.t. \begin{cases} Z = [a_{11}, a_{22}, \ldots, a_{nn}]^T \\ a_{ii} = \{0, 1\} \\ \sum_{i=1}^{n} a_{ii} = L \end{cases} \tag{20}$$

with optimization variables $\{a_{11}, a_{22}, \ldots, a_{nn}\}$. As can be seen, problem (27) is a standard 0–1 positive semi-definite quadratic programming problem which can be solved by many optimization toolboxes. Here, we adopt the *YALMIP* toolbox [20] to solve this problem.

Next, we will develop the optimization scheme to solve (21). The objective function of (20) can be equivalently rewritten as:

$$\min - tr\left(\left(K(X,X) - K(X,X)\left(K(X,X) + \delta^2 I\right)^{-1} K(X,X)\right)\Lambda\Lambda^T\right) \tag{21}$$

Obviously, objective function (28) is a linear function of $\{a_{ii}\}_{i=1,2,\ldots,n}$. Thus, the formulation of optimization problem (29) is the same as (27), where the matrix F is different.

$$\min_{a_{ii}} \frac{1}{2} Z^T GZ + Z^T F$$
$$s.t. \begin{cases} F = -\left[\sum_{i=1}^{k} u_i^T K_{c1} K_{c1}^T u_i \lambda_i \quad \cdots \quad \sum_{i=1}^{k} u_i^T K_{cn} K_{cn}^T u_i \lambda_i\right]^T \\ -\mu_2 R[1, 1\ldots., 1]_{1\times n}^T \\ E = \left(K(X,X) - K(X,X)\left(K(X,X) + \delta^2 I\right)^{-1} K(X,X)\right) \\ G = \mu_1 MM^T + NN^T \\ R = diag\{diag(E)\} \\ Z = [a_{11}, a_{22}, \ldots, a_{nn}]^T \\ \sum_{i=1}^{n} a_{ii} = L \\ a_{ii} = \{0, 1\} \end{cases} \tag{22}$$

Here, $R = diag\{diag(E)\}$ denotes that R is a diagonal matrix which is constructed by the diagonal entries of matrix E. As observed, problem (29) is also an 0–1 positive semi-definite quadratic programming problem, which can also be solved by the *YALMIP* toolbox.

4.2. The flow of AL-RU strategy

On the basis of the analysis given in previous subsections, the soft sensor modeling procedure based on *AL-RU* strategy is detailed as follows:

Step1: Initialize the candidate unlabeled sample pool X, the labeled sample dataset $X_L = \phi$ and the parameter $\varphi = \{\sigma_f, \sigma_n\}$ of the kernel function. Calculate the kernel matrix $K = K(X,X)$ and the centered kernel matrix $K_c = K - JK - KJ + JKJ$.
Step 2: Calculate the k highest eigenvalues $\lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_k\}$ and the corresponding eigenvectors $U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_k]$ of the centered kernel matrix K_c.
Step 3: The number of the selected samples at each iteration is set to L. Then, solve the optimization problem (16) by the *YALMIP* optimization toolbox and obtain the selected ID number of the most representative samples.
Step 4: Obtain their labels of the selected samples to form the updating dataset X_w. Update the labeled dataset $X_L = X_L + X_w$ and unlabeled dataset $X = X - X_w$. Then, use *GPR* to build the soft sensor model and update the parameter φ of the kernel function by the conjugate gradient algorithm. If the model performance satisfies the requirements, stop; otherwise, go to step5.
Step 5: Select another L informative samples and design the optimization problem as in (29). Then, solve it by the *YALMIP* optimization toolbox and obtain the new selected ID number. Go to step4.

A detailed illustration of the *AL-RU* soft sensor modeling is provided in Fig. 1.

5. Experimental results

To demonstrate the effectiveness of the proposed algorithm, two case studies of soft sensing are made to evaluate and compare five *AL* methods:

(1) **Random sampling** [14,15] (*RS*) which selects the samples randomly at each iteration.
(2) **A-optimal design** (*AOD*) [14] aims to minimize the trace of $\mathrm{cov}(\widehat{w}) = \mathrm{var}(X_s^T X_s)^{-1}$, where $\widehat{w}$ is the parameter of the linear regression model.
(3) ***AL-R*** adopts feature reconstruction method based on *KPCA* to select the representative samples, which is described in section 2.
(4) ***AL-U*** exploits the label information to select the uncertain samples, which is detailed in section 3.
(5) ***AL-RU*** selects the informative samples using different sampling criteria with the combination of feature reconstruction and uncertainty evaluation.

It is necessary to note that *RS*, *AOD*, *AL-R* algorithms are all label-independent. The *AL-U* algorithm selects the most uncertain samples based on the current soft sensor model, which is trained on some pregiven labeled samples. Here, the initial soft sensor model of *AL-U* algorithm is trained on the samples selected by random sampling, while the initial soft sensor model of *AL-RU* algorithm is obtained on the samples selected by *AL-R* sampling. To obtain stable results, *RS* and *AL-U* algorithms are repeated 10 times for each experimental test.

For evaluating the performance of the soft sensing models, the root mean square error (*RMSE*) criterion is defined as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \widehat{y}_i)^2}{N}} \quad i = 1, 2, \ldots, N \tag{23}$$

where N represents the number of testing samples, y_i and $\widehat{y}_i$ are the real and estimated values, respectively.

5.1. Debutanizer column

5.1.1. Process description and modeling preparation

The debutanizer column [21] is a conventional part of refinery process, which is always employed to illustrate the superiority of the soft sensing technology. The key variable, the butane concentration in the debutanizer bottoms, is measured by the gas chromatograph with a large measurement delay. Thus, a soft sensor for predicting the butane concentration in real time is necessary. According to process analysis, 7 indices (top temperature, top pressure, reflux flow, flow to next process, 6th tray temperature and two bottom temperatures) are chosen as the subsidiary variables which are easily measured by hardware sensors installed in the debutanizer column with the sampling rate of 15 min. The flowchart of the debutanizer column is shown in Fig. 2. For soft sensor modeling, 2000 data samples are collected and partitioned into two parts: 1000 samples are used as training dataset that contains unlabeled candidates whose labels are hidden deliberately for being queried, while the remaining 1000 samples serve as testing dataset. In order to test the performance of all the algorithms, 1% samples (10 samples) of the unlabeled candidates are assigned with the real labels and then added into

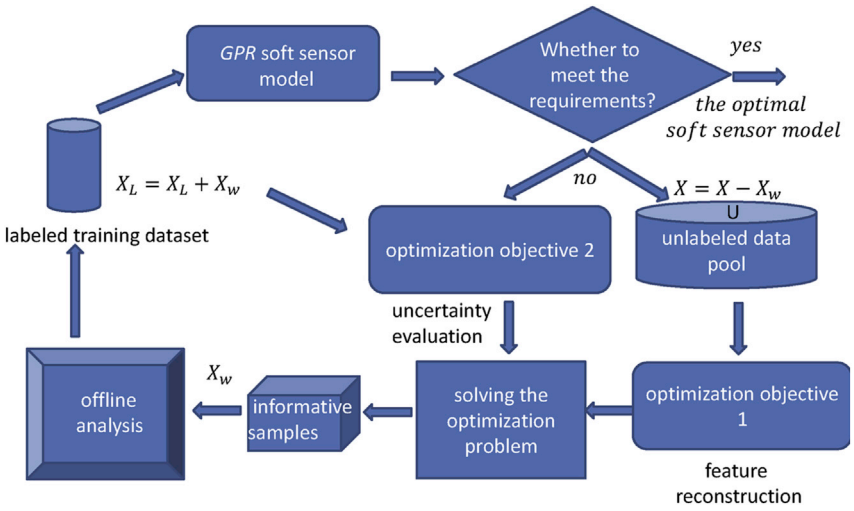


Fig. 1. The flowchart of *AL-RU* soft sensor modeling procedure.

the training dataset at each iteration. That means after 100 iterations, the unlabeled candidates will be exhausted. *GPR* modeling method is used as the training tool for soft sensing and the initial value of the kernel parameters is selected as $\varphi = \{1, 1\}$. Here, all the sample values are normalized to $[0, 1]$.

5.1.2. Experimental results and discussion

The proposed algorithm contains two regularization parameters: μ_1 and μ_2. Here, a 10-fold cross validation is implemented on the training dataset for parameters selection. For each dataset, we let *AL-R* algorithm select 50 training samples for modeling. Particularly, we apply *AL-RU* algorithm to iteratively select L $(= 10, 20, \ldots, 50)$ training samples for exploiting label information. We probe μ_1 in the range $[0.1, 1, 10, 100, 1000]$ and the range for μ_2 is $[0.1, 1, 10, 100, 1000]$. From the results of cross validation, it is found that regularization parameter $\mu_1 = 100$ best fit *AL-R* algorithm, and parameters $\mu_1 = 100$, $\mu_2 = 10$ get the best performance of *AL-RU* algorithm.

The comparison results of the *RMSE* values with the increasing of the labeled samples are shown in Fig. 3. The performance curve of *AL-RU* are fixed because the selected samples ID numbers at each iteration are calculated by optimization problem (29), while the performance curve of *RS* is not fixed due to the random sampling strategy. As a result, we choose the average performance curve to show the potential risk of *RS*. As observed, the soft sensor performance based on *AL* strategies is improved greatly during the first 50 iterations, while the performance improvement is limited at the 50–100 iterations. This is because the *AL* strategies adopt greedy algorithms which select the most informative samples at each iteration. Thus, with the increasing of the iterations, the additional information of the remaining candidates could hardly play a key role for the *GPR* model. However, *RS* has no sampling criterion to identify the most informative samples. As a result, the performance of *RS* is further improved in the last 50 iterations, which means *RS* needs more labeled samples to get a desired *GPR* model.

As can be seen, with the increasing of the labeled samples, *RMSE*

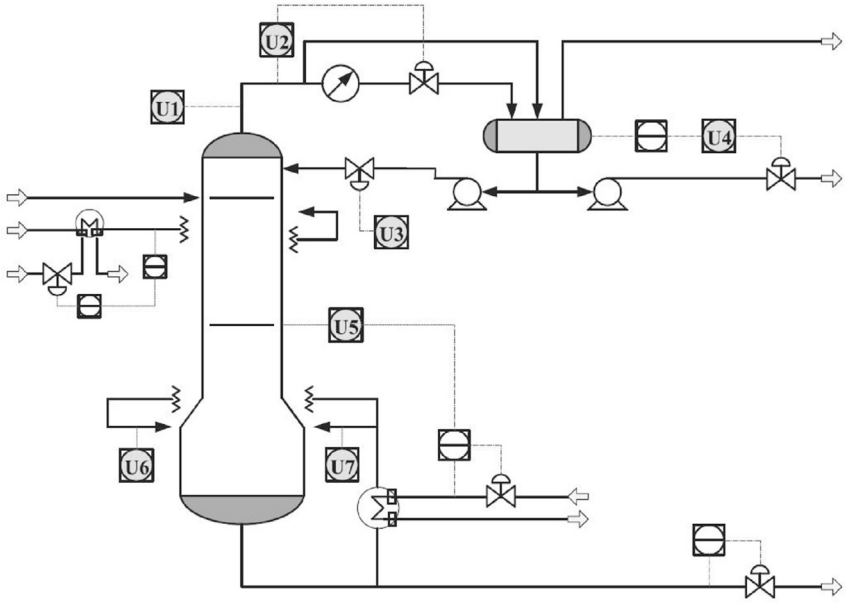


Fig. 2. The flowchart of the debutanizer column.

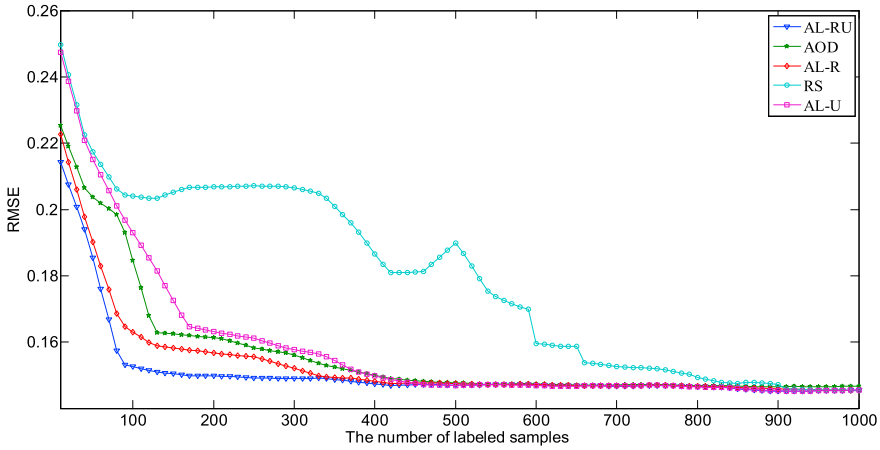


Fig. 3. Comparison of *RMSE* values for soft sensor models with different iterations.

values of most algorithms decrease except *RS*. If the candidate samples are selected by the random manner, the performance of the soft sensor may not be improved or even deteriorated. Because there is a potential risk that some bad quality samples, with measurement error and noise, may be selected for labeling and modeling. In addition, with incorporation of those randomly selected samples, the structure of the soft sensor model may be distorted.

It can be seen that the initial *RMSE* value of *AL-U* is larger than the other three *AL* strategies because *AL-U* randomly select the initial labeled samples at the first iteration. Meanwhile, the converge rate of *AL-U* is faster than *RS* algorithm due to the exploitation of the label information at the following iterations. Both *AOD* and *AL-R* adopt the unlabeled sampling criterion to select the most representative samples. However, *AL-R* performs better than *AOD*, which adopts *KPCA* to select the data samples that contain fewer noises and more principal components information. The converge speed of *AL-RU* is higher than the other algorithms which means that less labeled samples are needed to build a desired soft sensor. This is because *AL-RU* has a strong ability to select the most information samples by designing a trade-off between *AL-R* and *AL-U*.

Table 1
RMSE of soft sensors (test error + train error) for each algorithm.

L	*RMSE*(test error(train error))				
	Random sampling	*AOD*	*AL-R*	*AL-U*	*AL-RU*
10	0.2519(0.2362)	0.2267(0.2115)	0.2235(0.2007)	0.2513(0.2370)	**0.2167(0.1985)**
30	0.2344(0.2223)	0.2160(0.1955)	0.2064(0.1815)	0.2315(0.2108)	**0.2011(0.1896)**
50	0.2177(0.1946)	0.2029(0.1858)	0.1907(0.1807)	0.2172(0.1876)	**0.1836(0.1673)**
100	0.2034(0.1871)	0.1873(0.1691)	0.1621(0.1325)	0.1935(0.1786)	**0.1527(0.1254)**
200	0.2045(0.1892)	0.1621(0.1407)	0.1579(0.1270)	0.1633(0.1441)	**0.1451(0.1192)**
300	0.2035(0.1884)	0.1587(0.1344)	0.1565(0.1234)	0.1592(0.1384)	**0.1446(0.1151)**
500	0.1898(0.1669)	0.1454(0.1205)	0.1452(0.1189)	0.1451(0.1156)	**0.1441(0.1089)**

Bold represents the bset results with the same number of seleted samples *L*.

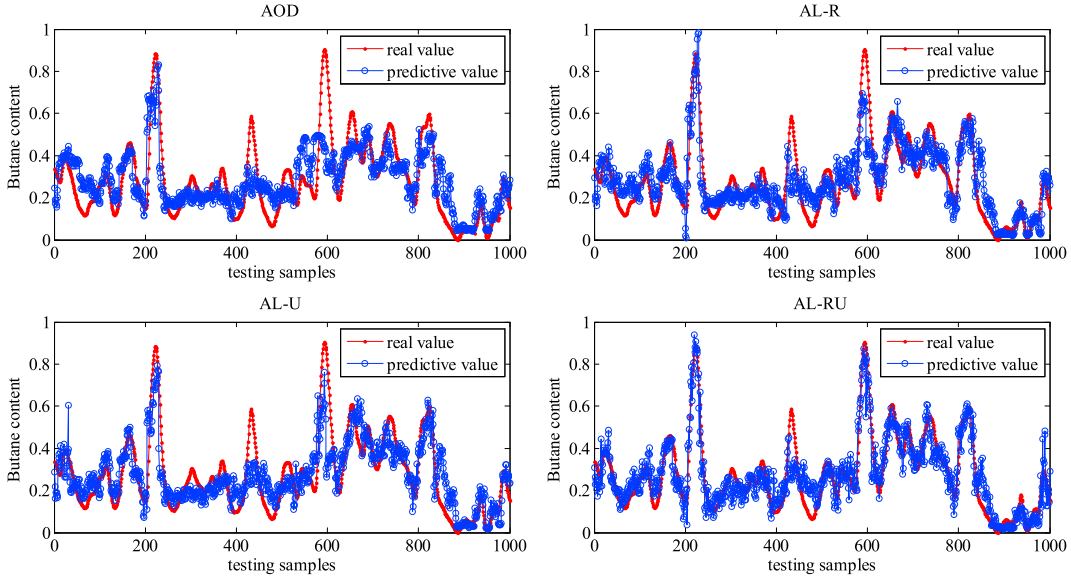


Fig. 4. The predictive results of butane concentration.

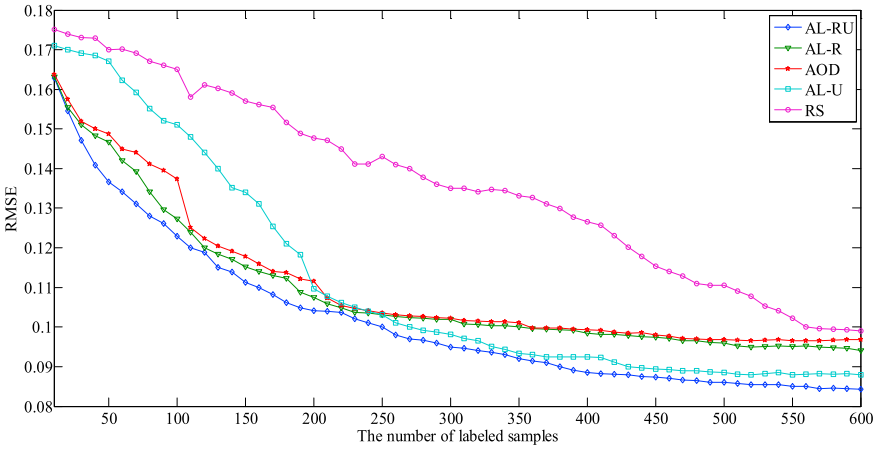


Fig. 5. Comparison of *RMSE* values for soft sensor models with different iterations.

Table 1 gives the results (training error and testing error) for each soft sensor modeling method. It can be seen that with the same number of selected samples, the *RMSE* values of *AL-RU* is smaller than that of the other algorithms. Particularly, with only 200 selected samples, *AL-RU* performs better than the other algorithms with 300 selected samples. Thus, *AL-RU* has a stronger ability to select the most informative samples compared to the other algorithms. The predictive results of butane concentration with 300 training samples are shown in Fig. 4. It can be observed that the predictive accuracy based on *AL-RU* is better than the other 3 *AL* algorithms. The *AL-RU* soft sensor model can successfully capture the data trend, especially for those samples between 580 and 620.

5.2. Circulating fluidized bed (CFB) boiler

5.2.1. Process description and modeling preparation

Circulating fluidized bed (*CFB*) boiler [22] is widely used for various industrial applications including cracking, drying, catalyst regeneration, power generation, and combustion, which has several advantages, such as fuel adaptability and lower emission of pollutants. One important factor of influencing the efficiency of *CFB* boiler directly is that some key variables are difficult to be measured online, such as flue gas oxygen content (*FGOC*) values that need offline analysis. Thus, a soft sensor for predicting *FGOC* in real time is necessary. Here, a practical application of 150t/h *CFB* boiler from Sichuan province in China is used to verify the proposed algorithm. According to the correlation analysis based on field data, 5 indices (coal feed rate, primary air flow, secondary air flow, flue gas temperature and steam flow) are chosen as the subsidiary variables. For soft sensor modeling, 700 data samples are collected with the sampling interval of 30 min. Then we divide the samples into two parts: 600 samples are used as training dataset that contains unlabeled candidates whose labels are hidden deliberately for the experiments, while the remaining 100 samples serve as testing dataset. We apply each algorithm to select 10 training samples for modeling at each step. Thus, the candidates will be exhausted after 60 iterations. *GPR* modeling method is used as the training tool for soft sensing and the initial value of the kernel parameters is selected as $\varphi = \{1, 1\}$. Here, all the sample values are normalized to $[0, 1]$.

5.2.2. Experimental results and discussion

On this dataset, we also use the 10-fold cross validation to select the appropriate parameters μ_1 and μ_2, where $\mu_1 \in [0.1, 1, 10, 100, 1000]$ and $\mu_2 \in [0.1, 1, 10, 100, 1000]$. From the results of the cross validation of *AL-R* and *AL-RU* algorithms, we obtain that regularization parameter $\mu_1 = 10$ best fit *AL-R* algorithm, and parameters $\mu_1 = 10$, $\mu_2 = 10$ get the best performance of *AL-U* algorithm.

The average results of the *RMSE* values with different number of labeled samples are shown in Fig. 5. Again, our *AL-RU* algorithm outperforms the other algorithms during all the iterations. *RS* performs worst, especially when there are less training samples due to the poor ability of capturing the informative samples. It can be seen that when the number of the selected samples is less than 200, *AL-R* and *AOD* performs better than *AL-U*; otherwise, When the number of the labeled samples is more than 200, *AL-U* performs better than *AL-R* and *AOD*. This is because during the initial several iterations, the label information is sparse so that the desired *GPR* model can not be built directly, which is not enough to make an appropriate sampling decision. However, with the increasing of the labeled samples, the updated *GPR* model becomes more and more reliable with the ascending ability of searching the informative samples. During the iterations, *AL-U* only calculates estimated variance for selecting the uncertain samples, but ignores the feature information of the candidates. As a result, *AL-U* performs worse than *AL-RU* which finds a reasonable trade-off between the representativeness and uncertainty.

Table 2 gives the results (training error and testing error) for each soft sensor modeling method. It can be seen that both training error and testing error of *AL-RU* are less than other algorithms. Thus, the *AL-RU*

Table 2
RMSE of soft sensors (test error + train error) for each algorithm.

L	*RMSE*(test error(train error))				
	Random sampling	*AOD*	*AL-R*	*AL-U*	*AL-RU*
100	0.1650(0.1137)	0.1373(0.0821)	0.1271(0.0784)	0.1510(0.0877)	**0.1229(0.0740)**
200	0.1477(0.0832)	0.1115(0.0714)	0.1075(0.0611)	0.1096(0.0703)	**0.1041(0.0573)**
300	0.1351(0.0811)	0.1022(0.0545)	0.1019(0.0578)	0.0981(0.0431)	**0.0950(0.0215)**
400	0.1266(0.0763)	0.0993(0.0443)	0.0985(0.0439)	0.0924(0.0117)	**0.0885(0.0106)**
500	0.1105(0.0639)	0.0969(0.0301)	0.0959(0.0217)	0.0885(0.0105)	**0.0860(0.0098)**
600	0.0990(0.0419)	0.0966(0.0245)	0.0940(0.0202)	0.0880(0.0101)	**0.0843(0.0083)**

Bold represents the bset results with the same number of seleted samples *L*.

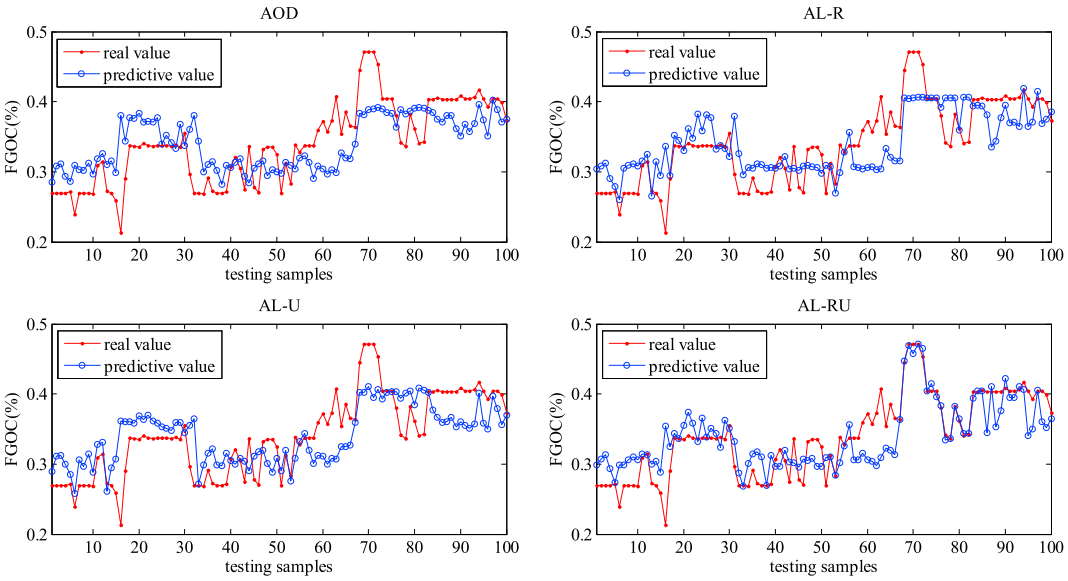


Fig. 6. The predictive results of *FGOC*.

algorithm has a stronger ability to exploit the information contained in the candidate samples. The predictive results of *FGOC* with 600 training samples (selected samples) are shown in Fig. 6. It can be observed that the performance of *AL-RU* soft sensor is better than the other 3 models. Particularly, for those samples between 71 and 78, the *AL-RU* soft sensor model can successfully capture the data trend.

6. Conclusion and future work

In this paper, we propose a new soft sensor modeling framework, named *AL-RU*, to select the informative samples to obtain their labels with the purpose of reducing human efforts. Comparing to other algorithms, such as *RS*, *AOD*, *AL-R* and *AL-U*, the proposed algorithm achieves the best performance. The experimental results demonstrate the effectiveness of the proposed method.

Central to our algorithm is the optimization scheme which contains two optimization objectives. The first is to select the most representative samples that have the best reconstruction of the global features, while the other is to choose the most uncertain samples by *GPR* model. A reasonable trade-off between the two sampling criteria are designed to improve the generalization ability of the soft sensor.

Although the *AL-RU* soft sensor can reduce human efforts, the performance will be degraded due to the changes in the process, including state change, abrasion of mechanic components and catalyst activity changes. To address this problem, lots of methods have been developed in the past few years, such as moving window methods, recursive adaptation methods, and ensemble methods. In future work, we plan to utilize these adaptive soft sensor modeling strategies to improve the proposed *AL* algorithm.

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